
CS689: Machine Learning

Fall 2024 - Midterm Exam: Solutions

Instructions: Write your name on the exam sheet. The duration of this exam is **two hours**. No electronic devices may be used during the exam. You may consult your paper notes and/or a print copy of texts during the exam. Sharing of notes/texts during the exam is strictly prohibited. Show your work for all derivation questions and explain your answers for all questions. Attempt all problems. Partial credit may be given for incorrect or incomplete answers. If you need extra space for answers, write on the back of the **preceding** page. If you have questions at any time, please raise your hand.

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1. (10 points) Loss Functions. The following problems concern loss functions for regression and classification.

a. (5 pts) Consider a regression problem where our goal is to penalize large errors more than the standard squared loss $L(y, y') = (y - y')^2$. Explain why we can not use the function $L(y, y') = (y - y')^3$ to accomplish this goal.

Example Solution: In order for loss functions to be well defined, it must be bounded below. The function $L(y, y') = (y - y')^3$ goes to negative infinity as y' goes to infinity for any fixed y . Thus, the function $L(y, y') = (y - y')^3$ is not bounded below and can not be used as a loss function.

b. (5 pts) Consider a binary classification problem with $y \in \{-1, 1\}$. The loss that we would like to use is the classification error function $L(y, f_\theta(\mathbf{x})) = [y \neq f_\theta(\mathbf{x})]$ where y is the true class label and $f_\theta(\mathbf{x})$ is the predicted class label, but this leads to a non-differentiable optimization problem. Does using the squared loss $L(y, f_\theta(\mathbf{x})) = (y - f_\theta(\mathbf{x}))^2$ fix this problem? Explain your answer.

Example Solution: Using the function $L(y, f_\theta(\mathbf{x})) = (y - f_\theta(\mathbf{x}))^2$ where $f_\theta(\mathbf{x})$ is a prediction function will not solve this problem. The issue is that the prediction function $f_\theta(\mathbf{x})$ must map the inputs $\mathbf{x}$ and parameters into the set $\{-1, 1\}$. Thus, $f_\theta(\mathbf{x})$ has to be a discontinuous function of the parameters and therefore $L(y, f_\theta(\mathbf{x})) = (y - f_\theta(\mathbf{x}))^2$ is still a non-differentiable function of the parameters.

2. (10 points) Empirical Risk Minimization. A constant classifier is a classifier that always outputs the same class regardless of the input. The prediction function is defined as $f_\theta(\mathbf{x}) = \theta$ where $\theta \in \{-1, 1\}$ is the only parameter. Use this model to answer the following questions.

a. (5 pts) Write down the empirical risk minimization problem for this model using the classification error function as the loss function and assuming a data set $\mathcal{D}$ containing N data cases. Briefly explain your answer.

Example Solution: The empirical risk minimization problem is to identify the parameters that minimize the risk. The risk is the average loss. The loss in this case is the classification error. Mathematically, we have:

$$\hat{\theta} = \arg \min_{\theta \in \{-1, 1\}} \frac{1}{N} \sum_{n=1}^N [y_n \neq f_\theta(\mathbf{x})] = \arg \min_{\theta \in \{-1, 1\}} \frac{1}{N} \sum_{n=1}^N [y_n \neq \theta]$$

b. (5 pts) What is the solution to this optimization problem? Explain your answer. (Hint: you can not use calculus to solve this problem).

Example Solution: Suppose we chose $\theta = 1$. The risk will be $\frac{1}{N} \sum_{n=1}^N [y_n \neq 1] = N^-/N$, the fraction of negative instances in the data set. Suppose we chose $\theta = -1$. The risk will be $\frac{1}{N} \sum_{n=1}^N [y_n \neq -1] = N^+/N$, the fraction of positive instances in the data set. Suppose that $N^+ \geq N^-$, then to minimize the loss we need to set $\theta = 1$. Otherwise, we should set $\theta = -1$. In other words, we should set θ to match the label of the class with the most instances in the training data set. This means we will make errors on the class with fewer examples in the training set.

3. (10 points) Closed-Form Optimization. Consider the objective function shown below for $x \in \mathbb{R}$. Find the global minimizers of this function. Show your work and explain briefly each step in your solutions.

$$f(x) = \frac{1}{4}x^4 - 8x^2$$

Example Solution: We first need to find the values of x where the derivative of $f(x) = 0$. These are the stationary points, which could be minima or non-minima. For a stationary point to be a minima, we need the second derivative of $f(x)$ to be positive. Since the function $f(x)$ is a polynomial and increases as x goes to $\pm\infty$, we know the function is bounded below. We proceed as follows:

$$f'(x) = \frac{d}{dx}\left(\frac{1}{4}x^4 - 8x^2\right) = x^3 - 16x$$

We now solve the stationary point equation:

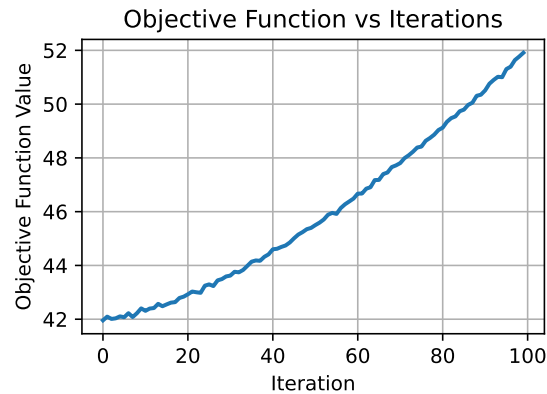
$$\begin{aligned}x^3 - 16x &= 0 \\ \Rightarrow x(x^2 - 16) &= 0 \\ \Rightarrow x = 0, x = \pm 4\end{aligned}$$

We now find the second derivative:

$$f''(x) = \frac{d^2}{dx^2}\left(\frac{1}{4}x^4 - 8x^2\right) = 3x^2 - 16$$

We have $f''(0) = -16$ and $f''(4) = 3(16) - 16 = 32$ and $f''(-4) = 3(16) - 16 = 32$. This means the local minima are $x = \pm 4$. We have $f(4) = f(-4)$, so both 4 and -4 are global minimizers.

4. (10 points) Numerical Optimization. Suppose we apply a steepest descent method from an optimization library to minimize an objective function of interest and obtain the following plot of the objective function value vs optimization iteration. Assume that the implementation of the optimization method, objective function and gradient computation are all correct. Consider this plot when answering the following questions.



a. (5 pts) What problem with the application of the optimization method does the above result indicate? Briefly explain your answer.

Example Solution: Since the loss is increasing and not decreasing and all functions are implemented correctly, this indicates that the learning rate is too large and the optimizer is diverging.

b. (5 pts) Describe two different ways that this problem could be fixed.

Example Solution: (1) We could use a step size that is sufficiently small so that the loss generally decreases and the optimizer eventually converges. (2) We can use backtracking line search with the Armijo rule to select the step size on every iteration. This will guarantee that the loss is non-increasing.

5. (10 points) Generalization Error. Suppose we are given a data set and are tasked to develop and learn a classification model. We use the data set to implement a standard train-test experiment design. Under what circumstances will the test error yield an unbiased estimate of the expected generalization error? Explain your answer.

Example Solution: If we are given a single data set and randomly partition it into a training set and test set using a standard train/test experiment design, the training data and the test data will follow the same distribution $p_*(\mathbf{x}, y)$. In order for the test error to be an unbiased estimate of the expected generalization error, we need the distribution of future data $p'_*(\mathbf{x}, y)$ and the distribution of the test instances to be the same. In other words, we need $p_*(\mathbf{x}, y) = p'_*(\mathbf{x}, y)$.

We also need to ensure that the test error is not used to select any hyper-parameters. Otherwise, even if $p_*(\mathbf{x}, y) = p'_*(\mathbf{x}, y)$, the test error of the model with the hyper-parameters that minimize the test error will be biased low relative to the expected generalization error.

6. (10 points) Convexity. The following questions concern convex functions and convex sets.

a. (5 pts) Consider the function $f(x) = \sqrt{x}$ where $x \in \mathbb{R}^{\geq 0}$. Use the definition of a convex function to prove whether this function is or is not convex.

Example Solution: A convex function must satisfy the condition $f((1 - \alpha)x + \alpha x') \leq (1 - \alpha)f(x) + \alpha f(x')$. This needs to hold for all $x, x' \geq 0$ and all $\alpha \in [0, 1]$. We will show that for $f(x) = \sqrt{x}$ this is not the case by counter example. Take $x = 0$ and $x' = 4$. Let $\alpha = 0.25$.

We have $f((1 - \alpha)x + \alpha x') = f((1 - 0.25)0 + 0.25(4)) = f(1) = \sqrt{1} = 1$.

On the other hand, we have $(1 - \alpha)f(x) + \alpha f(x') = (1 - 0.25)f(0) + 0.25f(4) = (1 - 0.25)\sqrt{0} + 0.25\sqrt{4} = 0 + 0.25(2) = 0.5$.

Since $f((1 - \alpha)x + \alpha x') > (1 - \alpha)f(x) + \alpha f(x')$, the function is not convex.

b. (5 pts) Let $\mathbf{w} \in \mathbb{R}^D$ be a fixed column vector. Consider the set of row vectors defined by $\mathcal{S} = \{\mathbf{x} | \mathbf{x} \in \mathbb{R}^D, \mathbf{x}\mathbf{w} \geq 0\}$. Use the definition of convexity to prove whether this set is or is not convex.

Example Solution: For $\mathcal{S}$ to be a convex set it must be the case that for all $\mathbf{x}$ and $\mathbf{x}'$ in $\mathcal{S}$ and $\alpha \in [0, 1]$, we have $\alpha\mathbf{x} + (1 - \alpha)\mathbf{x}' \in \mathcal{S}$. By assumption $\mathbf{x} \in \mathcal{S}$ so $\mathbf{x}\mathbf{w} \geq 0$. Since $\alpha > 0$ we have that $\alpha\mathbf{x}\mathbf{w} \geq 0$ since it is the product of two non-negative terms. Similarly, since $\mathbf{x}' \in \mathcal{S}$ we have $\mathbf{x}'\mathbf{w} \geq 0$. Further, since $\alpha \leq 1$ we have that $(1 - \alpha) \geq 0$ and $(1 - \alpha)\mathbf{x}'\mathbf{w} \geq 0$. This means that $\alpha\mathbf{x}\mathbf{w} + (1 - \alpha)\mathbf{x}'\mathbf{w} \geq 0$ since both terms are non-negative. Finally, by distributivity of matrix multiplication the previous result implies $(\alpha\mathbf{x} + (1 - \alpha)\mathbf{x}')\mathbf{w} \geq 0$ and thus $(\alpha\mathbf{x} + (1 - \alpha)\mathbf{x}') \in \mathcal{S}$. The set $\mathcal{S}$ is thus convex.

7. (10 points) Subdifferentials. Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined below. Derive the subdifferential function for this function. Show your work and explain each step of your answer.

$$f(x) = \begin{cases} 2x^2 - 5x + 4 & x < 1 \\ x^3 & x \geq 1 \end{cases}$$

Example Solution: This function has one potential point of non-differentiability at $x = 1$. For $x < 1$ the function is differentiable since it is a polynomial and similarly for $x > 1$ the function is differentiable since it is a polynomial. Thus, for $x \neq 1$, the subdifferential of $f(x)$ is the set containing the unique derivative of $f(x)$. At $x = 1$, the subdifferential is a closed interval where the lower bound is the derivative of the first piece of $f(x)$ evaluated at 1 and the upper bound is the derivative of the second piece of $f(x)$ evaluated at 1. To proceed, we find the derivatives of the two pieces, and then assemble the sub-differential function $\partial f(x)$:

$$\begin{aligned} \frac{d}{dx}(2x^2 - 5x + 4) &= 4x - 5 \\ \frac{d}{dx}(2x^2 - 5x + 4)|_{x=1} &= 4(1) - 5 = -1 \\ \frac{d}{dx}(x^3) &= 3x^2 \\ \frac{d}{dx}(3x)|_{x=1} &= 3(1) = 3 \\ \partial f(x) &= \begin{cases} \{4x - 5\} & \text{if } x < 1 \\ [-1, 3] & \text{if } x = 1 \\ \{3x^2\} & \text{if } x > 1 \end{cases} \end{aligned}$$

8. (10 points) Support Vector Machines. The following questions concern support vector machines.

a. (5 pts) Explain the primary advantage of linear support vector regression compared to OLS linear regression.

Example Solution: Linear support vector regression uses the epsilon insensitive loss while OLS linear regression uses the squared loss. Due to the fact that the epsilon insensitive loss increases only linearly as a function of $y - y'$ outside of the epsilon insensitive region, it is much less sensitive to outliers than the squared loss. This makes SVR more robust to outliers than OLS linear regression in the sense that outliers have much less of an effect on the learned model parameters.

b. (5 pts) Explain what problem is solved by the use of kernel functions instead of basis expansions when constructing non-linear support vector machines.

Example Solution: Using a basis expansion like the polynomial expansion causes the size of the basis expanded feature space to grow exponentially in D as the polynomial order increases. The polynomial kernel solves this problem by enabling the computation of the inner product between two basis expanded vectors without needing to explicitly compute the polynomial expansion. Depending on the value of N , D and the polynomial order, the use of the kernel can require both less storage space and less compute time when learning the model.

9. (10 points) Neural Networks. The following questions concern neural networks.

a. (5 pts) Explain the primary advantage of using a neural network instead of a basis expanded linear model when designing a non-linear classification model.

Example Solution: The primary advantage of neural networks over basis expanded linear models is that a neural network can learn feature representations from data while the basis expanded linear model uses a fixed feature representation. This gives neural networks lower bias than the basis expansion when a sufficient number of hidden units is used.

b. (5 pts) Explain one disadvantage of using a neural network instead of a basis expanded linear model when designing a non-linear classification model.

Example Solution: There are two primary disadvantages. First, neural networks have higher variance than basis expanded linear models when a large number of hidden units is used. This means they can need more data for learning to perform well compared to a basis expansion.

Second, the learning problem for neural networks has many local optima. Basis expanded linear models including OLS linear regression, logistic regression and support vector machines all have convex optimization problems with unique global minima.

10. (10 points) Implementation. Suppose you are trying to develop a Numpy implementation of a probabilistic classification model for a problem where $\mathbf{x} \in \mathbb{R}^D$ and $y \in \{-1, 1\}$. You find the following two implementations of the negative log likelihood function on Stack Overflow. Which should you use? Explain your answer.

```
def nll(y_true, probs):
    n = len(y_true)
    loss = 0.0
    for i in range(n):
        if(y_true[i]>0):
            loss -= np.log(probs[i])
        else:
            loss -= np.log(1 - probs[i])
    return loss

def nll(y, p):
    return -1*np.sum(y*np.log(p) + (1-y)*np.log(1-p))
```

Example Solution 1: Since the labels in the data set are $y \in \{-1, 1\}$, we can not use the second function since it is the form of the binary cross entropy that expects $y \in \{0, 1\}$. While the first implementation is not vectorized, it is correct for labels in $y \in \{-1, 1\}$. Given the choice between a vectorized function that is incorrect and a non-vectorized function that is correct, we should pick the function that is correct.

Example Solution 2: Since the labels in the data set are $y \in \{-1, 1\}$, we can not use the second function as is since it is the form of the binary cross entropy that expects $y \in \{0, 1\}$. However, we can easily map our -1 label values to 0 and then use the second function. This would be preferable since the second function is vectorized and is otherwise correct for $y \in \{0, 1\}$.